

Big data project presentation

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Completed tasks so far

- **T0: Data preparation**
- **T1: Data processing**
- **T2: Data quality**
- **T3: Exploratory data analysis**
- **T5: Data augmentation**
- **T6: Streaming**
- **T8: FHVHV market impact**

T0: Data preparation (BRONZE layer)

1. Get missing parquet files

Arnes HPC cluster

missing_data.py

2. Upload to Snowflake BRONZE stage



snowflake.py

Tables 5

- T0_FHVVHV_SCHEMA
- T0_FHV_SCHEMA
- T0_GREEN_SCHEMA
- T0_SCHEMA_ALL
- T0_YELLOW_SCHEMA

3. Tables with info about column names and data types

Stages 4

- FHVHV_RAW_STAGE
- FHV_RAW_STAGE
- GREEN_RAW_STAGE
- YELLOW_RAW_STAGE

T1: Data processing (SILVER layer)

BRONZE

Tables 5

- T0_FHVVHV_SCHEMA
- T0_FHV_SCHEMA
- T0_GREEN_SCHEMA
- T0_SCHEMA_ALL
- T0_YELLOW_SCHEMA

- Schema unification
- Standardize column names and data types
- Add pickup_year, pickup_month

SILVER

Tables 4

- FHVHV_TRIPS
- FHV_TRIPS
- GREEN_TRIPS
- YELLOW_TRIPS

Preview: BIGDATA_TAXI_MZMB.SILVER.YELLOW...

Table Chart Pivot													100 rows 955ms	
#	VENDOR_ID	PICKUP_DATETIME	DROPOFF_DATETIME	PASSENGER_COUNT	TRIP_DISTANCE	RATECODE_ID	STORE_AND_FWD_FLAG	FU_LOCATION_ID	DO_LOCATION_ID	PAYMENT_TYPE	FARE_AMOUNT	EXTRA		
16	2	2013-12-28 09:49:00.000	2013-12-28 09:54:00.000	4	1.39	1	null	48	237	2	6.5	0		
17	1	2013-12-03 06:38:27.000	2013-12-03 06:44:09.000	1	1.1	1	N	79	113	1	6.5	0		
18	1	2013-12-09 14:09:08.000	2013-12-09 14:19:41.000	1	2	1	N	186	50	1	9	0		
19	2	2013-12-28 20:19:00.000	2013-12-28 20:33:00.000	1	2.05	1	null	48	234	1	10.5	0.5		
20	1	2013-12-11 20:56:23.000	2013-12-11 21:09:24.000	3	1.9	1	N	158	144	2	10.5	0.5		
21	1	2013-12-29 12:21:24.000	2013-12-29 12:26:28.000	1	0.6	1	N	230	50	2	5.5	0		
22	1	2013-12-10 05:17:37.000	2013-12-10 05:29:29.000	1	4.3	1	N	163	7	2	14.5	0.5		
23	1	2013-12-30 13:28:54.000	2013-12-30 13:46:06.000	1	2.4	1	N	79	186	2	13.5	0		
24	1	2013-12-11 20:56:43.000	2013-12-11 21:44:05.000	1	11.2	1	N	107	243	1	40.5	0.5		
25	1	2013-12-03 08:03:35.000	2013-12-03 08:22:38.000	1	12.5	5	N	125	1	1	76	0		
26	1	2013-12-04 10:20:41.000	2013-12-04 10:41:59.000	1	11.4	1	Y	138	164	2	33	0		
27	1	2013-12-03 01:54:14.000	2013-12-03 02:17:06.000	2	6.4	1	N	79	36	2	22	0.5		
28	2	2013-12-28 09:17:00.000	2013-12-28 09:22:00.000	1	1.39	1	null	68	48	1	6	0		
29	2	2013-12-01 04:32:00.000	2013-12-01 04:49:00.000	1	5.61	1	null	79	65	1	19	0.5		
30	1	2013-12-14 16:39:57.000	2013-12-14 16:49:14.000	4	0.7	1	N	79	148	1	7	0		
31	2	2013-12-01 23:46:00.000	2013-12-01 23:52:00.000	1	1.28	1	null	113	231	2	6.5	0.5		
32	1	2013-12-11 20:22:46.000	2013-12-11 20:28:39.000	1	0.8	1	N	142	143	1	5.5	0.5		

T2: Data quality (GOLD layer summaries)

✓ SILVER

✓ Tables 4

- FHVHV_TRIPS
- FHV_TRIPS
- GREEN_TRIPS
- YELLOW_TRIPS

-
- invalid pickup year
 - pickup outside source month
 - dropoff before pickup
 - zero/nonpositive distance
 - negative fare/amount
 - invalid passenger count

✓ GOLD

✓ Tables 54

- T2_FHVHV_QUALITY_CHART
- T2_FHVHV_QUALITY_FILE_YEAR
- T2_FHVHV_QUALITY_YEAR
- T2_FHV_QUALITY_CHART
- T2_FHV_QUALITY_FILE_YEAR
- T2_FHV_QUALITY_YEAR
- T2_GREEN_QUALITY_CHART
- T2_GREEN_QUALITY_FILE_YEAR
- T2_GREEN_QUALITY_YEAR
- T2_YELLOW_QUALITY_CHART
- T2_YELLOW_QUALITY_FILE_YEAR
- T2_YELLOW_QUALITY_YEAR

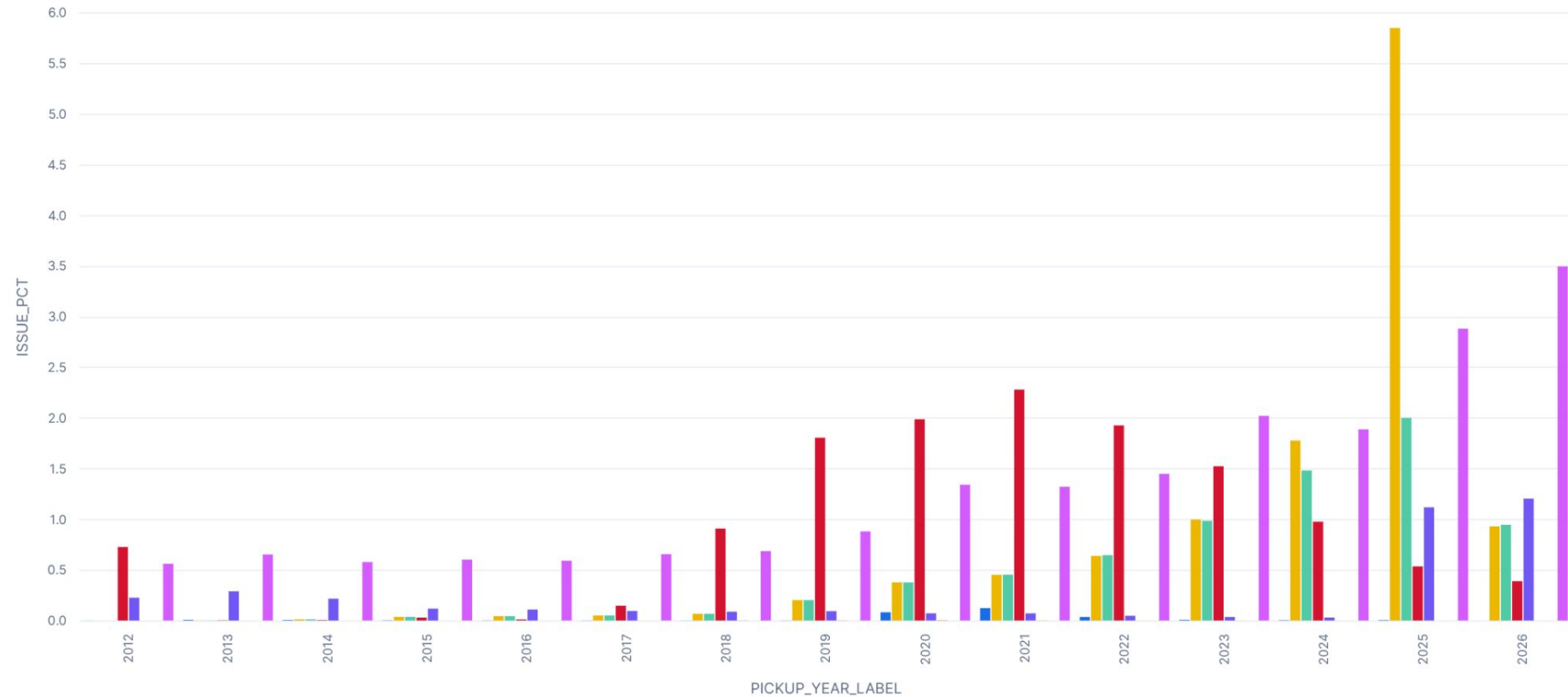
✓ Views 4

- FHVHV_TRIPS_CLEAN
- FHV_TRIPS_CLEAN
- GREEN_TRIPS_CLEAN
- YELLOW_TRIPS_CLEAN

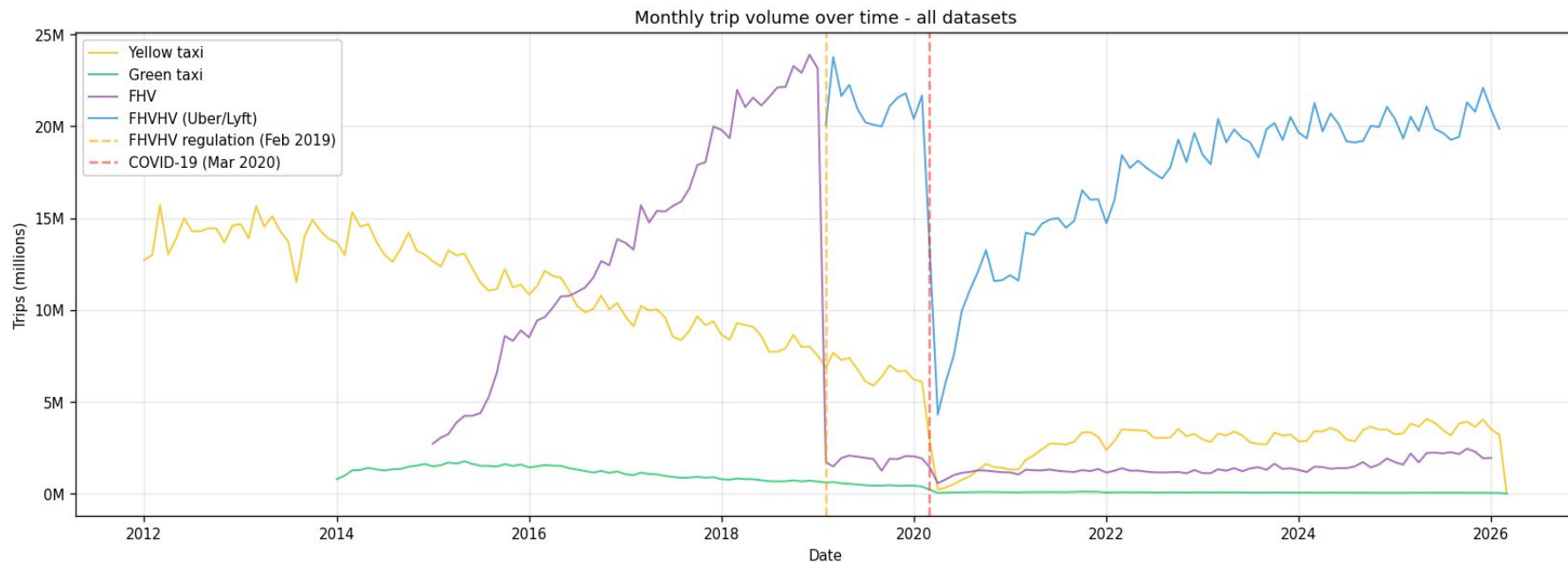
T2: Data quality (GOLD layer)

ISSUE_TYPE

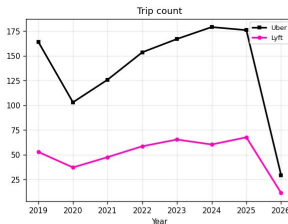
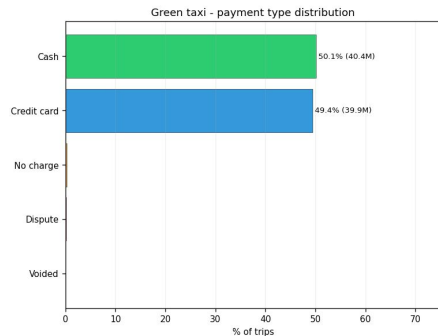
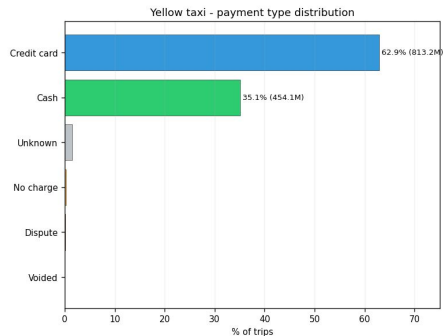
dropoff_before_pickup negative_fare_amount negative_total_amount nonpositive_passenger_count pickup_equals_dropoff pickup_outside_source_month unexpected_pickup_year zero_trip_distance



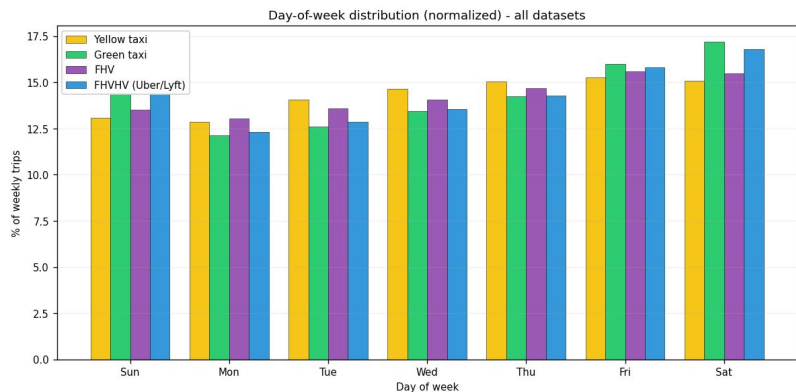
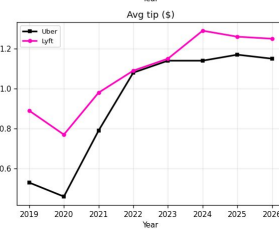
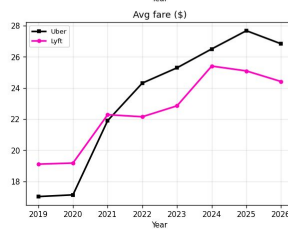
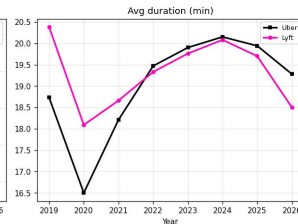
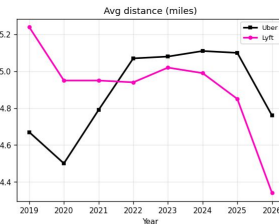
T3: Exploratory data analysis



T3: Exploratory data analysis



Uber vs Lyft comparison over time



T5: Data augmentation

EXTERNAL_DATA

Tables 10

- T5_ATTRACTIONS_BY_ZONE
- T5_ATTRACTIONS_RAW
- T5_EVENTS
- T5_EVENTS_HOURLY
- T5_SCHOOLS_BY_ZONE
- T5_SCHOOLS_RAW
- T5_WEATHER
- TAXI_ZONES
- TAXI_ZONES_GEOM
- TAXI_ZONE_LOOKUP

Table	Source	Content
T5_WEATHER	Open-Meteo archive API	Daily max/min temp (°C), precipitation (mm) for NYC, 2012–2025
T5_SCHOOLS_RAW → T5_SCHOOLS_BY_ZONE	NYC Open Data (n3p6-zve2)	Primary and high schools with lat/lon, spatially joined to taxi zones
T5_ATTRACTIONS_RAW → T5_ATTRACTIONS_BY_ZONE	NYC Open Data (fn6f-htvv)	Museums and cultural institutions, spatially joined to taxi zones
T5_EVENTS → T5_EVENTS_HOURLY	NYC Open Data (bkfu-528j)	Permitted public events 2020–2025 by borough and datetime

Column	Description
BOROUGH, ZONE, SERVICE_ZONE	From TAXI_ZONE_LOOKUP by pickup location
TEMP_MAX_C, TEMP_MIN_C	Daily temperature at pickup date
PRECIPITATION_MM	Daily precipitation at pickup date
SCHOOL_COUNT	Number of schools in the pickup zone
ATTRACTION_COUNT	Number of attractions in the pickup zone
ACTIVE_EVENTS	Number of permitted events active during the pickup hour in the pickup borough

T6: Streaming

Combined Yellow +
FHVHV trips for 2021

 T6_STREAM_SOURCE_2021

Replay ordered trips
into Kafka

 producer_t6_2021.py

Kafka topics:
t6-yellow-2021 /
t6-fhvhv-2021



ingested messages into GOLD

 T6_KAFKA_LANDING_2021

Snowflake Kafka
connector

Rolling 24h stats: boroughs +
top 10 locations

 T6_ROLLING_BOROUGH_STATS_2021

 T6_ROLLING_TOP_LOCATION_STATS_2021

 consumer_t6_birch_2021.py

BIRCH clustering algorithm = builds a
clustering-feature tree and clusters
rows

T8: FHVHV market impact

